Jobs Calculator

The Jobs Calculator calculates the net employment change needed to achieve a target unemployment rate after a specified number of months. The user can adjust the target unemployment rate, the number of months, and the assumed labor force participation rate.

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Set assumptions

Choose a geography, review the latest data, then adjust the target inputs.

Geography

United States

Current statistics: United States

Last Update: October 2, 2026
Next Update: November 6, 2026

Unemployment Rate:	**4.2%**
Monthly Employment Change:	**+29,000**
Labor Force Participation Rate:	**61.8%**

Target inputs

Target unemployment rate

4.2 %

Time to target

12 months

Assumed labor force participation rate

61.8 %

Underlying assumptions

Population Growth (monthly)

0.043 % / month

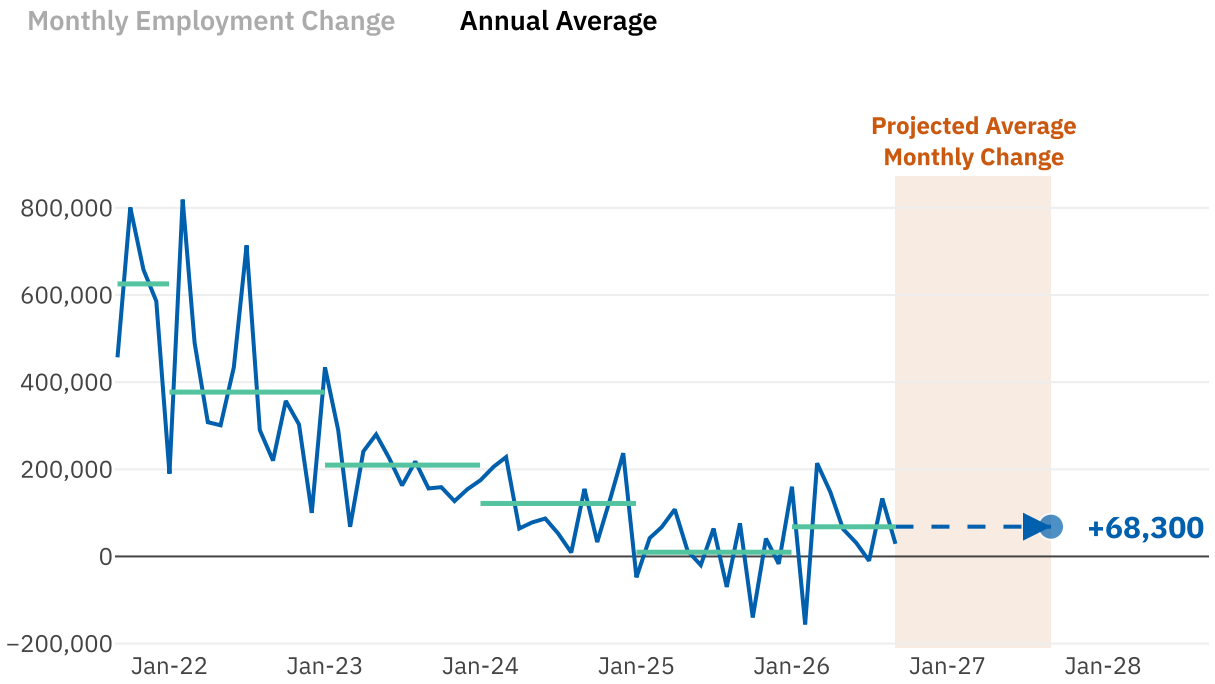
CES/CPS Ratio

97.4 %

Calculate | Reset | Compare scenarios

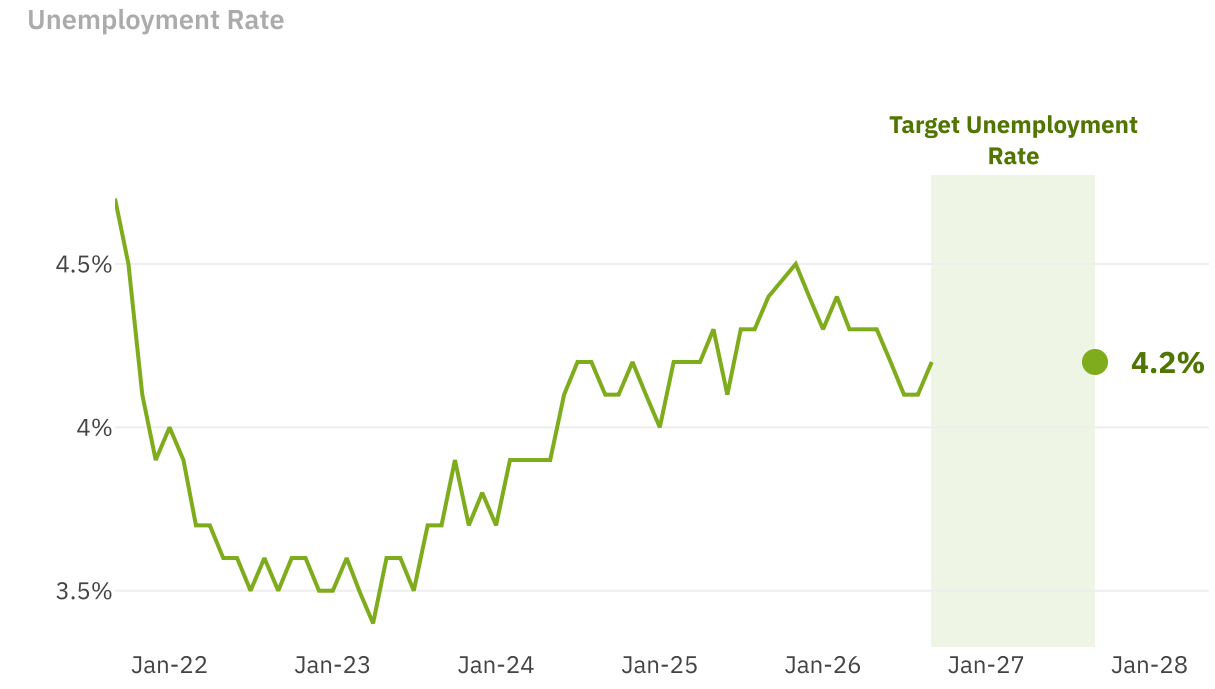
Date range for charts Last 5 years

The United States will need **+68,300** jobs per month on average over the next **12** months ...



Source: US Bureau of Labor Statistics Summary Table B, Establishment data. Atlanta Fed Calculations.

Based on projected unemployment rate over the next **12** months ...



Source: US Bureau of Labor Statistics Summary Table A, Household Data

And the average labor force participation rate is **61.8%**.



1. Why does the Federal Reserve care about the unemployment rate?

Section 2A of the Federal Reserve Act states, "The Board of Governors of the Federal Reserve System and the Federal Open Market Committee shall maintain long-run growth of the monetary and credit aggregates commensurate with the economy's long run potential to increase production, so as to promote effectively the goals of maximum employment, stable prices, and moderate long-term interest rates." This part of the Federal Reserve Act is often referred to as the Fed's "dual mandate." Basically, it states that the Federal Reserve's monetary policy has the goals of stable prices and maximum employment. The unemployment rate is the most popular statistic that measures the degree to which the Federal Reserve has achieved the goal of maximum employment.

2. What are the sources of the numbers behind the Jobs Calculator?

The current labor market statistics reported each month by the U.S. Bureau of Labor Statistics (BLS) are used to support the Jobs Calculator. These statistics are derived from two surveys, the Household Survey and the Establishment, or Payroll, Survey . (For definitions of terms and concepts, consult the official BLS Handbook of Methods .) Of course, these sources supply the underlying statistics; the user supplies the desired unemployment rate and time frame.

3. Why does household employment differ from payroll employment?

The Establishment (or Payroll) Survey estimates the number of jobs for which a paycheck was written in the United States during a particular pay period, whereas the Household Survey is a measure of individual employment status. For example, if one person holds two jobs, the Payroll Survey will count two jobs, but the Household Survey counts one person employed.
The difference arises mostly because the scope of the Household Survey is broader than the Payroll Survey. The Household Survey includes the self-employed, unpaid family workers, agricultural workers, and private household workers—all of these workers are excluded from the sample frame for the Payroll Survey.

The two employment numbers typically follow the same cyclical paths, although the gap has varied from time to time. (See research such as Juhn and Potter 1999 and Wu 2004 for further information.)

4. Does a decline in the labor force participation rate mean people are leaving the labor force because they are discouraged about finding a job?

According to the BLS definition, a discouraged worker is someone who wants a job, is available to work, has sought work sometime in the previous 12 months, but has not looked for work in the previous four weeks because of a belief that no jobs for which they would qualify are available. On average, 93 percent of those not in the labor force in 2011 said that they did not want a job so they would not have been eligible to be classified as discouraged. Among those who were not currently in the labor force, but who wanted a job and had looked for work in the previous 12 months, 38 percent were classified as "discouraged," 8 percent were not in the labor force because of family responsibilities, 13 percent were in school or undergoing training, 6 percent indicated health or disability for their reason to not be in the labor force, and 34 percent were out of the labor force for some other unspecified reason.

5. How important is labor force growth in hitting the target unemployment rate anyway?

It's very important. The labor force and the unemployment rate together determine the new level of employment (see the details for how the Jobs Calculator works), as shown here:
New Employment = (1 – New Unemployment Rate) x New Labor Force

The number of months necessary to achieve a given unemployment rate is also important. For example, try using the Jobs Calculator to change only the number of months. Leave everything else the same and notice what happens to the implied monthly growth in the labor force. The longer the time period you allow—keeping the labor force participation rate and population growth constant—the smaller the implied growth in the labor force and consequently, the fewer number of net jobs need to be created each month to absorb the implied change in the labor force.

6. What about other measures of unemployment?

The Jobs Calculator uses the definition of unemployment and labor force most commonly highlighted in the media and economic research (see the BLS's explanation). The BLS also computes five other unemployment rate measures using different definitions of the labor force; for example, including discouraged job seekers. In general, these various unemployment rate measures tend to move together over time.

7. Where can I find more information on the unemployment rate?

The BLS website is a great resource. See, for example, "How the Government Measures Unemployment " or the FAQs about labor force statistics.
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How Many Jobs? Details behind the Jobs Calculator

The number of jobs an economy supports, the number of people who want jobs but don't have one, the number of people willing to work...these are all factors that define the state of the labor market. The purpose of this page is to explain what is behind the Jobs Calculator.

On the first Friday of every month, the U.S. Bureau of Labor Statistics (BLS) reports statistics that define the condition of the U.S. labor market. The two statistics superstars—the ones that get the most attention—are the payroll employment numbers and the unemployment rate. The question that seems be on everyone's mind is, how many new jobs does the economy need to create each month to reduce the unemployment rate to some specific level? The purpose of the Jobs Calculator is to allow the user to answer that very question, based on certain assumptions (some of which the user can adjust).

The payroll employment numbers and the unemployment rate are estimated by the BLS from responses to two separate surveys.

Establishment Survey

Through the Current Employment Statistics (CES) program, the BLS surveys approximately 141,000 nonfarm businesses, covering about 486,000 worksites, asking employers about employment, hours, and earnings of their workers. This survey of establishments is commonly referred to as the Payroll Survey.

The total employment number that is reported from the Payroll Survey reflects an estimate of the number of people in the United States who received a paycheck for work during the pay period that includes the 12th day of the month. This count is considered to most accurately give the total number of jobs in the United States at a given point in time.

Household Survey

The BLS also surveys about 60,000 households every month to obtain estimates of employment and nonemployment activity, total income, and demographics of the population of the United States (those 16 years and older). The reference period for activity is the same week as the

Payroll Survey: the week that includes the 12th day of the month. This survey is called the Current Population Survey (CPS).

From responses to the Household Survey, the BLS calculates the total labor force, the labor force participation rate, and the unemployment rate. While the user should consult the official BLS Handbook of Methods for complete definitions of these terms, here are the basic concepts and formulas:

Employed (E): A person who has a job for pay, is temporarily absent from work, or works unpaid in a family business.

Unemployed (U): A person who does not have a job, is available to work, and has looked for work during the previous four weeks.

Labor Force (LF): The sum of everyone who is employed and unemployed (restricted to noninstitutionalized civilians).

Labor Force Participation Rate (LFPR): The labor force divided by the population (POP), $LFPR = LF/POP$.

Unemployment Rate (UR): The number unemployed divided by the labor force, $UR = U/LF$.

How the Jobs Calculator works

As the formulas above show, in order to calculate how many new jobs are needed to move the UR from one number to another, one needs to make an assumption about the growth in the labor force. The labor force can grow as people enter or return to the labor market (for example, people graduating high school or college and looking for a first job, or new immigrants looking for work upon arrival to the United States). The labor force can also shrink based on people leaving the labor force (for example, to retire, return to school, or care for household members). It's not easy to predict how changes in demographics or behavior will change the size of the labor force (see several papers on this subject in the additional reading section below).

As an example, suppose that total employment in one month (period 1) is E_1 and the unemployment rate in the same month is UR_1. Suppose we want to know how many new jobs (E_2-E_1) will be needed in order to reduce the unemployment rate to some level in period 2, UR_2. This number is determined by the following formula:

E_2-E_1 = [(1-UR_2)*LF_2]-E_1.

This formula is derived by the following manipulation of the formula for the unemployment rate: $UR_2 = (U_2/LF_2) = (LF_2 - E_2)/LF_2$. Solving for E_2 and subtracting the first period's employment level, E_1, produces the formula above.

It's clear that an assumption about what the labor force looks like in period 2 is crucial; the greater the increase in the labor force, the greater the increase in employment required to attain a particular unemployment rate, UR_2. The Jobs Calculator allows the user to investigate the impact a different growth rate in the labor force has on the number of jobs that are needed to attain a certain unemployment rate by changing assumptions about the labor force participation rate and the monthly population growth rate. The higher the labor force participation rate (for a given population growth), the greater the growth in the labor force. The higher the population growth rate (for a given labor force participation), the greater the growth in the labor force.

Translating the number of jobs from the Household Survey to the Payroll Survey:

One might expect that the employment estimate produced by the Payroll and Household Surveys would be the same number. Theoretically, they both claim to estimate the number of jobs in the economy. However, they are different for several reasons.

The Payroll Survey estimates the number of jobs for which a paycheck was written in the United States during a particular pay period, whereas the Household Survey is a measure of employment *activity*. For example, if one person holds two jobs, the Payroll Survey will count the two jobs, but the Household Survey counts one person employed.

In addition—and the main source of the difference—is that the scope of the Household Survey is broader than the Payroll Survey. The Household Survey includes the self-employed, unpaid family workers, agricultural workers, and private household workers—all of these workers are excluded from the establishment survey.

Since the headline jobs number refers to the Payroll Survey and the unemployment rate comes from the Household Survey, the Jobs Calculator makes a simple transformation to the household employment calculation to produce the estimate of the number of payroll jobs needed.

Default assumptions

The Jobs Calculator makes three assumptions, two of which the user can adjust. The default monthly population growth is set so that it equals the average monthly population growth during the previous 12 months (excluding months that contain Census population adjustments). The default labor force participation rate is set so that it equals the current month's labor force

participation rate. The default CES/CPS multiplier (which is not adjustable by the user) is the current month's ratio of payroll employment to household employment.

Additional reading

Abraham, Katherine G.; John Haltiwanger; Kristin Sandusky; and James R. Spletzer. "Exploring Differences in Employment between Household and Establishment Data ." *Journal of Labor Economics* 31(2) (April 2013): S129-S172.

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Hall, Robert E. "Labor Market Frictions and Employment Fluctuations ." NBER Working Paper no. 6501 (April 1998).

Hotchkiss, Julie L. "Changes in the Aggregate Labor Force Participation Rate." *Federal Reserve Bank of Atlanta Economic Review* 94(4) 2009: 1–6.

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Juhn, Chinhui, and Simon Potter. "Explaining the Recent Divergence in Payroll and Household Employment Growth ." Federal Reserve Bank of New York *Current Issues in Economics and Finance* 5(16) (December 1999): 1–6.

Wu, Tao. "Two Measures of Employment: How Different Are They? " *Federal Reserve Bank of San Francisco Economic Letter* (August 27, 2004).

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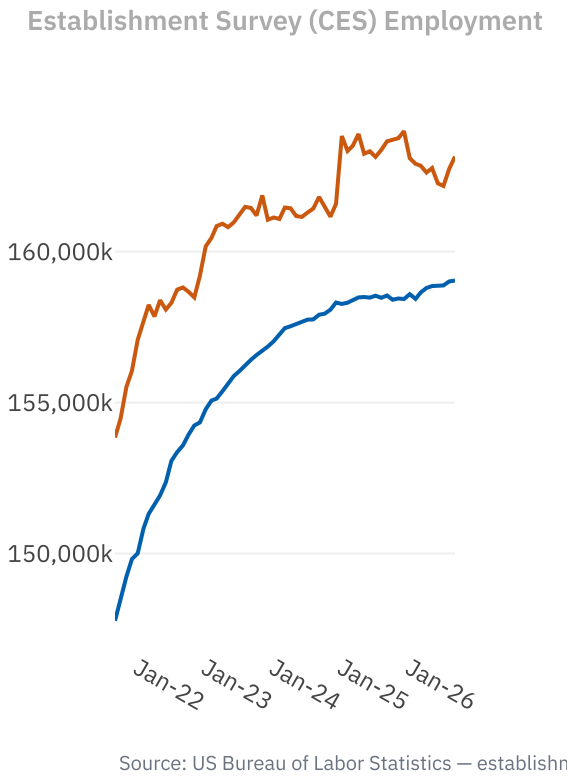
Below are historical data that underlay the Jobs Calculator for reference.

Geography

United States

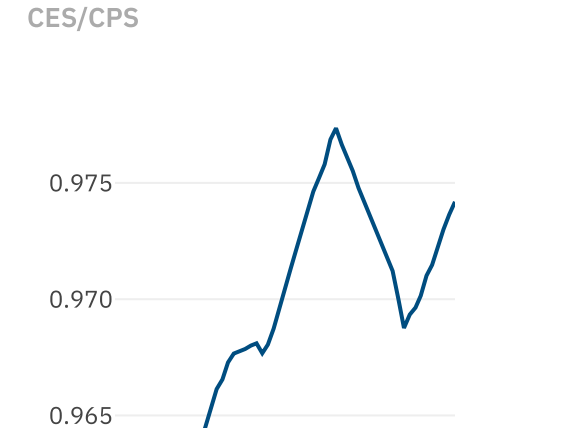
Date range for charts Last 5 years

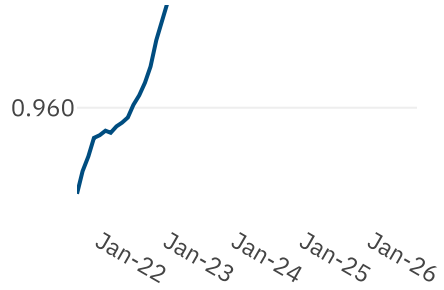
Establishment versus Household Survey Employment Levels | thousands of persons, seasonally adjusted



Source: US Bureau of Labor Statistics — establishn

Establishment-to-Household Survey Employment Ratio (CES/CPS) | 12-month moving average, seasonally adjusted

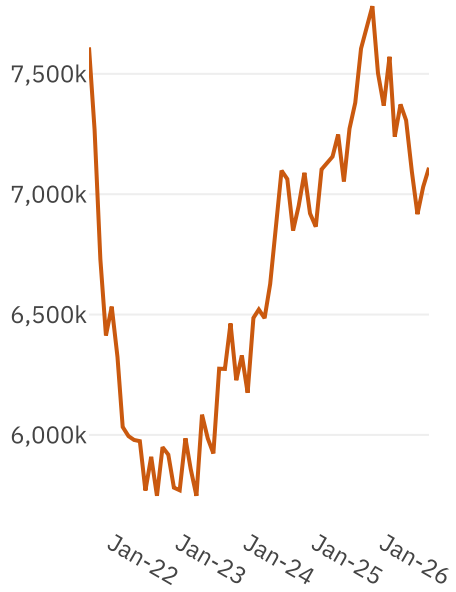




Source: US Bureau of Labor Statistics. 12-month n

Unemployment Level | thousands of persons, seasonally adjusted

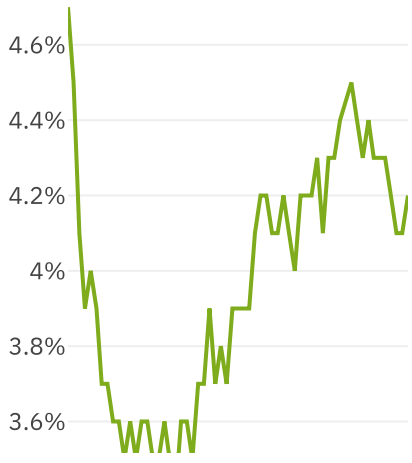
No. of Unemployed



Source: US Bureau of Labor Statistics. Civilian labc

Unemployment Rate | percent, seasonally adjusted

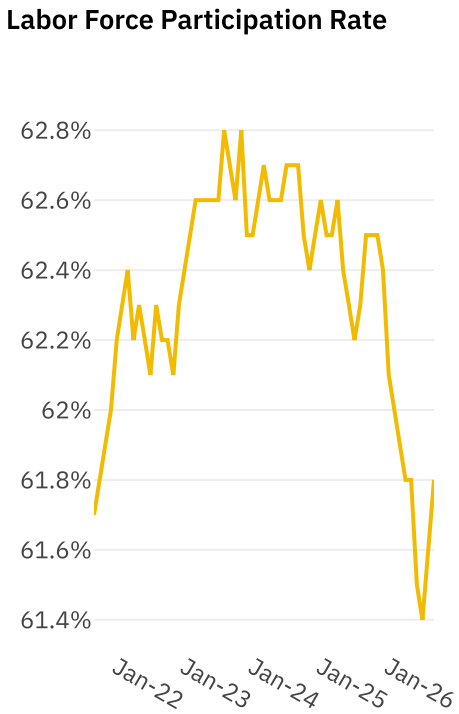
Unemployment Rate





Source: US Bureau of Labor Statistics. Percent, sea

Labor Force Participation Rate | percent, seasonally adjusted



Source: US Bureau of Labor Statistics. Percent, sea

Monthly Population Growth Rate | percent, 12-month moving average

